Skills Summary

Algebraic Expressions — Distributing and Combining Safely

Use this reference while completing your worksheet or when studying.

Skill 1 — Distribute to every term, sign included

Rule: $k(p+q) = kp + kq$ — the multiplier outside reaches *every* term inside, and it keeps its own sign while doing it.

Signs: negative times negative is positive, so $-2(x-3) = -2x + 6$. Write the sign as part of the number you are distributing, and the second term takes care of itself.

The substitution test: put any number in both versions. Equivalent expressions always agree; one disagreement proves the step was illegal.

Example: Write $6(x-4)$ without parentheses.
Step 1. Two terms sit inside the parentheses, so distributing must produce two products — the subtraction sign belongs to the second one.
Step 2. $6 \cdot x = 6x$ and $6 \cdot (-4) = -24$, and at $x = 5$ both versions give 6, which is the check.
$\boldsymbol{6x - 24}$

Try it: Write $-2(x+7)$ without parentheses.

check: $\boldsymbol{-2x - 14}$

Skill 2 — Combine only like terms

Like terms have exactly the same letter part: $7a$ and $-2a$ are like; $5x$ and $3y$ are not, and $5a$ and 13 are not.

Rule: add or subtract the coefficients and keep the letter part unchanged — $7a - 2a = 5a$, never $5a^2$ and never 5.

Unlike terms simply stay put. $5x + 3y$ is already simplified; an expression is allowed to have more than one term, and forcing it into one is the most common error on this page.

Example: Simplify $4b + 7 - b + 2$.
Step 1. Sort the terms into groups with the same letter part first, carrying each sign with its own term, so nothing unlike ever meets.
Step 2. The b terms give $4b - b = 3b$; the constants give $7 + 2 = 9$; and $3b$ and 9 are unlike, so the work stops there. $\boldsymbol{3b + 9}$

Try it: Simplify $9m - 3 - 4m + 8$.

check: $\boldsymbol{5m + 5}$

Skill 3 — Factor out, then distribute back

Factoring is distributing in reverse: find the greatest common factor of the terms, write it outside, and write what is left inside.

$kp + kq = k(p + q)$. Divide EVERY term by the common factor — leaving one term undivided is the same mistake as distributing to only one term.

Always check: distribute your answer back. If it does not rebuild the original expression exactly, the factoring is wrong and you will see it at once.

Example: Factor $8x + 20$.
Step 1. Both terms share a numerical factor, so the greatest common factor comes out front and each term is divided by it.
Step 2. The common factor is 4; dividing gives $8x \div 4 = 2x$ and $20 \div 4 = 5$, and distributing back returns $8x + 20$ ✓. $\mathbf{4(2x + 5)}$

Try it: Factor $9x + 12$, then distribute back to check.

check: $\mathbf{3(3x + 4)}$

(!) Watch out: the two errors are opposites. Distributing to only the first term does too little; merging unlike terms does too much. One substitution catches either one in about ten seconds.